

Disproof of Conjecture 00000009650

Zero cross-overlaps in the real symmetric class

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Abstract

For the source's definition $o_{ij} = \langle u_i, u_j \rangle^2$ using eigenvectors of distinct eigenvalues of the same matrix, all such overlaps of a real symmetric matrix are zero. In particular the GOE case named in the source has a point-mass distribution, zero mean, and zero scaled mean. This contradicts the claimed density and universal $2/n$ scaling.

The elementary obstruction

We use the real symmetric class alone, so no interpretation of a square of a complex inner product is needed. Let T be real symmetric and suppose $Tu_i = \lambda_i u_i$ and $Tu_j = \lambda_j u_j$, where $\lambda_i \neq \lambda_j$. Self-adjointness of T gives

$$\lambda_i \langle u_i, u_j \rangle = \langle Tu_i, u_j \rangle = \langle u_i, Tu_j \rangle = \lambda_j \langle u_i, u_j \rangle.$$

Thus $(\lambda_i - \lambda_j) \langle u_i, u_j \rangle = 0$ and

$$o_{ij} = \langle u_i, u_j \rangle^2 = 0.$$

The argument is exact; it holds for every such matrix and pair. In particular it applies to every distinct-eigenvalue pair of every GOE realization, since a GOE matrix is real symmetric.

For a simple-spectrum matrix of size $n \geq 2$, set

$$P_n = \{(i, j) : 1 \leq i < j \leq n\}, \quad m_n = |P_n| = \binom{n}{2}.$$

The strict upper triangle is the relevant one for the source's explicit "distinct eigenvalues" requirement. Its empirical measure and mean are

$$\mu_n = \frac{1}{m_n} \sum_{(i,j) \in P_n} \delta_{o_{ij}} = \delta_0, \quad \bar{o}_n = \frac{1}{m_n} \sum_{(i,j) \in P_n} o_{ij} = 0.$$

More strongly, for *every* test function $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$\frac{1}{m_n} \sum_{(i,j) \in P_n} f(o_{ij}) = f(0).$$

If eigenvalues repeat and only the distinct-eigenvalue pairs are retained, each retained overlap is still zero. GOE has simple spectrum almost surely: repeated eigenvalues are the zero set of the discriminant polynomial, this polynomial is nonzero (evaluate it on a diagonal matrix with distinct entries), and GOE has a density on the space of real symmetric matrices. Consequently the same empirical law is δ_0 almost surely.

Proposition 1. *The claimed law in Conjecture 00000009650 fails in the real symmetric class.*

Proof. For all $n \geq 2$, the mean is $0 \neq 2/n$. If the mean assertion is intended as the asymptotic equivalent $\bar{o}_n \sim 2/n$, it also fails, since $n\bar{o}_n = 0$ for every n and cannot tend to 2. Furthermore $\mu_n = \delta_0$ for every n , so its limit is δ_0 . A probability measure with a Lebesgue density assigns zero mass to the singleton $\{0\}$, whereas $\delta_0(\{0\}) = 1$. Hence this limit cannot have the asserted density, with exponential tails or otherwise. $\square$

Scope and interpretation

The counterargument uses the same-matrix inner products printed in both language versions, not overlaps between eigenvectors of independent matrices or non-Hermitian left/right eigenvector condition numbers. The Bourgade–Dubach paper mentioned in the source concerns nonnormal Ginibre matrices [1]; it does not change the elementary identity above.

If one includes diagonal entries despite the distinct-eigenvalue wording, normalized eigenvectors give n ones and $n(n-1)/2$ zeros in the inclusive upper triangle. The empirical measure is then

$$\frac{n-1}{n+1}\delta_0 + \frac{2}{n+1}\delta_1 \implies \delta_0.$$

Its mean is asymptotic to $2/n$, but its limit still has no Lebesgue density. This optional convention does not rescue the complete conjecture.

Formalization and reproducibility

The Lean project proves the orthogonality identity over actual real inner product spaces. It defines the finite set of strict upper-triangle pairs, the stated overlaps and empirical test-function averages, and proves the point-mass identity for every test function. It derives the exact mean contradiction and the failure of convergence of the scaled mean to 2 for *any sequence* of real symmetric simple-spectrum eigensystems.

To exhibit actual objects as well, it constructs, for every size $n \geq 2$, $T_n = \text{diag}(0, 1, \dots, n-1)$ with the standard unit basis. Lean checks symmetry, distinct eigenvalues, normalization, the eigenvector equations and the violated mean formula. These diagonal examples are not presented as random GOE samples: the general orthogonality theorem is what applies to GOE. The probability-zero discriminant observation, elementary measure interpretation and inclusive-diagonal remark above are explained in this manuscript; the formal contradiction already follows from the pointwise mean theorems. No random-matrix universality theorem is assumed.

The project pins Lean 4.19.0 and Mathlib v4.19.0 at commit

c44e0c8ee63ca166450922a373c7409c5d26b00b.

Reproduction commands, successful build logs, foundational axiom outputs, file hashes and a separate adversarial self-review accompany the submission. The Python script performs exact rational finite checks as a supplement; the all-dimension proofs and real limits are in Lean.

References

- [1] P. Bourgade and G. Dubach, *The distribution of overlaps between eigenvectors of Ginibre matrices*, Probability Theory and Related Fields 177 (2020), 397–464. [arXiv:1801.01219](https://arxiv.org/abs/1801.01219).